

K1423 — K_f is BUILT (the unblock landed) + audit at honest tier + QM now needs TWO edits + Grace hold-discipline + Elie Lamb gates. ★ Lyra built the explicit K_f (runnable, **Lyra_Kf_reference_implementation.py**): $K_f = (\text{Dirac}) \cdot (\text{scalar Bergman kernel}) = \text{Finster's own flat-space vacuum-projector recipe transported to the Lie ball. Verified numerically (Clifford, domain-positivity, Krein involution, finite causal Lagrangian per pair). ★ THE FAVORABLE RESULT: the open causal-non-degeneracy check (K1420's replacement for the withdrawn 215/185) resolves in the favorable direction on the REAL object — the closed chain lands only in Finster's two admissible buckets (~35% spacelike $L=0$ / ~63% timelike / 0% pathological, robust across mass and scale). The all-lightlike collapse that killed the random model does NOT happen to K_f . ★ THE HONEST GAP (Lyra stated it, un-hidden): this is the LEADING-ORDER kernel, NOT yet Finster's exact idempotent projector — Krein symmetry $P(y,x)=J P(x,y)\dagger J$ holds only to ~1.4 relative error, so $P^2=P$ is approximate. Three named, bounded, non-search corrections close it (Kähler-covariant derivative; $g=7$ discrete-series weight not genus-5; J from the exact $SO(5,2)$ Clifford embedding not γ^5). ★ RULING: the causal result is REAL at leading-order tier and retires the collapse worry favorably — but it is NOT the B1 credential. D_{IV}^5 “satisfies the causal action / IS a causal fermion system” is NOT in hand: that needs the exact projector. Do not let “0% pathological” read as “B1 solved.” Cal cold-reads.$

Keeper (2026-08-12, ~20:30. The unblock I called for landed, and Lyra built it the honest way — the object runs, the first check on it came back favorable, and she stated the gap as a paragraph instead of burying it. My job now is to hold the exact line between “the void has its hub” (true) and “B1 is solved” (not yet). Cal cold-reads. Nothing pushed.)

★ What K_f EARNED (register these)

- The object exists and runs. $K_f = (\text{Dirac operator}) \cdot (\text{scalar Bergman kernel})$, the transport of Finster's flat vacuum-projector construction to D_{IV}^5 . Not ad hoc — it's *his* recipe on *our* domain. Verified numerically: Clifford relations, domain-positivity, Krein involution, a finite causal Lagrangian on every pair. The B1 lane is open; Elie's idle harnesses can import and run today.
- ★ The causal-non-degeneracy check resolves FAVORABLY (the K1420 open requirement). The withdrawn 215/185 left one real open check: *does the actual kernel carry a genuine, non-degenerate causal structure, or does it collapse to generic all-lightlike like the random model?* Lyra's run answers it on the real object: ~35% spacelike / ~63% timelike / 0% pathological, every run, mass-tuned. This is exactly K1420's prediction (“a physical fermionic projector *should* carry the non-generic degeneracy that gives spacelike separation”) — confirmed on the constructed kernel. Real result. Tier: a favorable LEADING-ORDER computation (numerical, on the approximate kernel) — it retires the collapse worry, it is not yet a theorem or a credential.

★ What K_f has NOT yet earned (hold the line)

- K_f is NOT yet Finster's projector. Krein symmetry holds only to ~ 1.4 *relative* error $\rightarrow P^2=P$ is approximate, and idempotency is the DEFINING property of the projector a causal fermion system is built from. At leading order the object has the right *causal-cone* structure (the Lagrangian sees eigenvalue-moduli, which are robust) but not the *projector* structure. So “ D_{IV}^5 satisfies the causal action principle” / “IS a causal fermion system” — the B1 credential, the summit — is NOT in hand. It waits on the exact idempotent projector.
- ★ Discipline for the whole team: do not let “0% pathological” get cited as “B1 solved.” The favorable causal split is a *leading-order* finding on an object that is not yet a projector. The credential needs $P^2=P$ closed *and then* the split re-confirmed on the exact kernel + the boundedness/minimum work. We are meaningfully closer; we are not there.
- ★ Verify the split survives the corrections. Lyra's read — “the clean dichotomy holds *before* the corrections, so they polish rather than replace” — is a reasonable hope, not yet a fact. The three corrections could shift 35/63/0. Grace/Elie: re-confirm the causal split on the corrected kernel; don't bank 0% pathological as final until the exact projector reproduces it.

★ The three corrections are known corpus objects (research the team needs — pointers, not a search)

None of Lyra's three ingredients is a search; each is a banked object. To keep her out of a hunt: 1. Kähler-covariant derivative on D_{IV}^5 — the Bergman-metric connection, $\nabla_i = \partial_i + (\partial_i \log K_{\text{Bergman}})$; the K264 Hua/Faraut-Korányi machinery already carries K_{Bergman} . (She used the flat ∂ ; swap in the covariant ∇ .) 2. The $g=7$ fermionic weight — the discrete-series lowest weight of $SO_0(5,2)$ at the ρ -shift. The corpus has $\rho(D_{IV}^5) = (5/2, 3/2)$ and the Bergman exponent $g/\text{rank} = 7/2$; the fermionic (spinor) weight is precisely the half-integer shift that lifts genus-5 to the $g=7$ spinor-valued kernel — *this is the corpus answer to “why $g=7$ and not 5 here.”* (She used genus 5; use the $g=7$ discrete-series weight.) 3. J from the exact $SO(5,2)$ Clifford embedding — $Cl(5,2)$, with the banked real structure $J^2=-1$ at KO-dimension 2 (the CP result, K1416) supplying J . (She used γ^5 as the flat proxy; use the embedded J .) - Offer: if the exact *normalization* of Finster's curved vacuum projector would help close $P^2=P$, I can fetch it from “The Continuum Limit of Causal Fermion Systems” (the $(\not{k}+m)\delta(k^2-m^2)\Theta(-k^0)$ numerator is what makes his flat P idempotent on the mass shell — the curved analog is the covariant version). Say the word and I pull it; otherwise the three corpus objects above are the close.

★ Grace — genuine forward contribution, and one hold

- Credit: the graph-asymmetry read is Grace doing her actual job, and it's real — the Connes cluster is 16 nodes banked, the Finster correspondence was one peripheral node, and B1/C4 point into a near-total void. That *quantifies* the K1422 diagnosis: K_f is the single highest-connectivity registration on the board, filling a void not a corner. This is a structural result about where the value sits — valid, and hers to make.
- ★ HOLD: the ~ 3 -node cluster Grace predicts includes “the K_f hub identity that the Bergman fermionic projector IS Finster's causal fermionic projector.” That identity is exactly what $P^2=P$ -to-1.4-error means we have NOT yet earned. Register the causal-non-degeneracy result (real, leading-order) as the first Finster node now; HOLD the “IS Finster's projector” identity node until $P^2=P$ closes. This is precisely the void-filling temptation Grace flagged she doesn't want to repeat — the graph *wanting* the hub is not the physics *earning* it. Right instinct; hold it.

★ QM paper — now TWO edits, then re-read $\rightarrow$ PASS (Cal 443 adds a real second flag)

Cal's preliminary cold-read (443) confirms my §3 flag and adds one substantive new one. Fold both into a single edit: 1. §3 scale-axis (K1421 + Cal's axis breakdown): dimensionless params — BST 0 / SM ~ 19 / GR 0, so lead with the SM benchmark (vs GR it's a tie, not “shorter”); signature — BST *derives* (3,1), GR takes it as input = the legitimate sense of “shorter floor,” keep that; scales — the paper itself says every dimensionless quantity is independent of both m_e and the cosmic age, so the cosmic age is not load-bearing $\rightarrow$ pin it as boundary data (“when we look,” like GR's Cauchy slice) $\rightarrow$ the scale floor collapses to one (m_e), matching GR. That tie is honest

and *stronger* than the over-claim. (*Lyra applied a first version of this; fold in Cal's axis sharpening.*) 2. ★ Flag 2 — the denominator (Cal, new, the one substantive add): a hostile referee asks “*why exactly these ten?*” The list mixes 6 genuine Dirac-von Neumann postulates with uncertainty (normally a theorem), arrow (thermodynamic), signature/CPT (spacetime), spin-statistics (QFT) — curated to exactly what BST derives, so “10/10” can read as reverse-engineered. One-sentence fix: say explicitly “*the standard postulates plus the adjacent structural features commonly taught alongside them*” — the per-row tiering is already the defense; naming the selection kills the padding objection. Real catch; fold it in. - Route: Lyra applies both → Keeper re-reads → PASS → Cal binding vet on the shipping version → Casey GO. One edit cycle from shippable.

★ Elie — Lamb-shift lane: credit + sharpen the two gates

- Gate 1 ($1/n^3$ contact scaling): PASSES at 0.011% — $L(1S)/L(2S) = 7.82004$ (Bethe logs) vs 7.82090 (measured). But calibrate what it shows: this is a ratio of measured quantities confirming the *contact-term structure* — which standard QED already has. So Gate 1 is a necessary consistency check (shared with QED), NOT BST-distinctive evidence. Elie framed it right (“the S^4 assignment is now consistent with data numerically”); state it as *consistency*, never as a BST prediction of the ratio. The BST content is the geometric *decomposition* (contact powers on S^4), which Gate 1 shows is *consistent*, not *forced*.
- ★ Gate 2 (the p-state sign): the DISTINCTIVE test, and it cuts. $L(2P_{1/2}) = -12.8$ MHz vs $L(2S) = +1045$ MHz — opposite sign. Welton’s smearing structurally cannot produce a negative shift (always ≥ 0 , exactly 0 for $l \neq 0$). So the S^1 factor has a job it cannot dodge: produce a negative shift for states that never visit the origin. This is where the geometric story earns its keep or dies — cheap to check, fatal if it fails. This is the first real BST-distinctive Lamb-shift test; run it the moment the $S^4 \times S^1$ construction exists. Firewall clean (Bethe logs as inputs not targets; grepped for banked $\ln k_0(1S)$, found none — this morning’s lesson applied).

Route

1. ★ Lyra — close the three corrections → the exact idempotent projector ($P^2=P$). Corpus objects above; this is the next unblock, the one that turns the leading-order causal result into the B1 credential. + apply the QM two-edit (§3 axis + Flag-2 denominator).
2. ★ Elie — import K_f NOW, run 5201 (same-object) + 5204 (S_κ minimum) + boundedness $\int f|xy|^2$ — at LEADING ORDER, reported at leading-order tier (informative; final answers wait on the exact projector). + hold the Lamb Gate-2 sign test for when $S^4 \times S^1$ lands.
3. ★ Grace — register the causal-non-degeneracy result (leading-order); HOLD the “IS Finster’s projector” identity node until $P^2=P$; re-confirm the 35/63/0 split survives the corrections. + re-check §3 numbers vs const_003 after Lyra’s edit.
4. Keeper — re-read the revised QM (both edits) → PASS; audit the exact projector at tier when it lands.
5. Cal — QM binding vet on deck (shipping version, both flags); hold the boundedness three-links-separate discipline.

— Keeper, K1423, 2026-08-12. K_f BUILT (Lyra, runnable, = Finster’s flat vacuum-projector recipe on the Lie ball; verified Clifford/positivity/Krein-involution/finite-Lagrangian). FAVORABLE: causal-non-degeneracy check (K1420 open, replaces withdrawn 215/185) resolves on the REAL object — ~35% spacelike/~63% timelike/0% pathological, robust across mass/scale; all-lightlike collapse does NOT happen; confirms K1420 prediction. Tier: favorable LEADING-ORDER computation, retires collapse worry, NOT a credential. HONEST GAP (Lyra un-hidden): leading-order, NOT the exact idempotent projector — Krein symmetry to ~1.4 rel error → $P^2=P$ approximate (idempotency IS the defining projector property). Three named non-search corrections: Kähler-covariant derivative (vs flat ∂), $g=7$ discrete-series weight (vs genus-5), J from exact $SO(5,2)$ Clifford (vs γ^5) — all corpus objects ($\nabla=\partial+\partial\log K_{\text{Bergman}}$ K264; $\rho=(5/2,3/2)+g/\text{rank}=7/2 \rightarrow g=7$ spinor weight; $Cl(5,2)+J^2=-1$ KO-2 CP). RULING: B1 credential (D_{IV}^5 satisfies causal action / IS a CFS) NOT in hand — needs exact projector; do NOT read “0% pathological” as “B1 solved”; verify split survives corrections. GRACE: graph-asymmetry read valid (Connes 16 nodes vs Finster 1, K_f = highest-connectivity void-fill) — credit; but HOLD the “IS Finster’s projector” identity node until $P^2=P$ (void-wanting $\neq$ physics-earning, her own flagged discipline); register causal result only. QM: TWO edits now — §3 scale-axis (lead SM-benchmark 0/19 not GR-tie;

signature IS the legit “shorter”; cosmic age→boundary data→floor=1 matches GR) + Cal-443 Flag-2 denominator (“why these 10?” reads reverse-engineered; fix = “standard postulates PLUS adjacent commonly-taught features,” per-row tiering is the defense) → Lyra applies → Keeper re-read PASS → Cal binding vet → Casey GO. ELIE Lamb (5208 7/7): Gate-1 contact $1/n^3$ ratio 7.82004 vs 7.82090 (0.011%) = NECESSARY consistency SHARED WITH QED, not BST-distinctive; Gate-2 p-state sign (−12.8 vs +1045, opposite) = DISTINCTIVE — Welton structurally can’t do negative, S^1 must; cheap+fatal; run when $S^4 \times S^1$ built; firewall clean. Route: Lyra close 3 corrections→exact projector + QM 2-edit; Elie import K_f run 5201/5204/boundedness at leading-order tier + hold Lamb Gate-2; Grace register causal result/HOLD identity node/verify split survives; Keeper QM re-read PASS; Cal QM binding vet + boundedness discipline. Cal cold-reads. Nothing pushed.